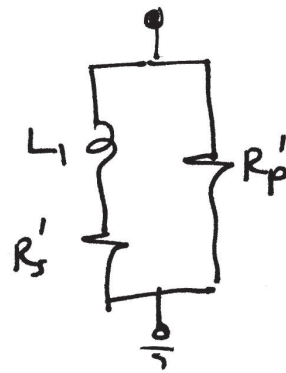


Problem 7.2 & 7.4:

Fig 7.32(a)

$$Y_1(s) = \frac{1}{R'_s + sL_1} + \frac{1}{R'_p}$$



$$Z_1(s) = \frac{1}{Y_1(s)} = \frac{R'_p (R'_s + sL_1)}{R'_p + R'_s + sL_1}$$

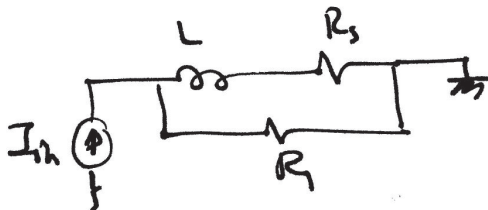
$$Z_1(j\omega) = \frac{R'_p (R'_s + j\omega L_1) (R'_p + R'_s - j\omega L_1)}{(R'_p + R'_s)^2 + \omega^2 L_1^2}$$

$$= R'_p \frac{R'_s (R'_p + R'_s) + \omega^2 L_1^2 + (R'_p + R'_s - R'_s) j\omega L_1}{(R'_p + R'_s)^2 + \omega^2 L_1^2}$$

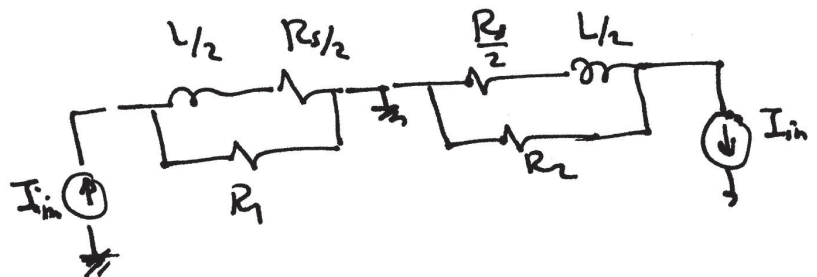
$$Q = \frac{\text{Im}(Z'_1)}{\text{Re}(Z'_1)} = \frac{R'_p \omega L_1}{R'_s (R'_s + R'_p) + \omega^2 L_1^2} \quad \text{eq (7.62)}$$

$$\text{if } R'_s \rightarrow 0 \Rightarrow Q = \frac{R'_p}{\omega L_1}, \quad \text{if } R'_p \rightarrow \infty, \quad Q = \frac{\omega L_1}{R'_s}$$

Using 7.62



7.37(b)



7.37(d)

7.37(b):

$$Q_b = \frac{R_1 \omega L}{R_s (R_s + R_1) + \omega^2 L^2}$$

7.37(d):

$$Q_d = \frac{R_1 \omega (L/2)}{(R_s/2) \left[\frac{R_s}{2} + R_1 \right] + \frac{\omega^2 L^2}{4}}$$

$$= \frac{(2 R_1) \omega L}{R_s [R_s + 2 R_1] + \omega^2 L^2}$$

if Q is dominant by R_s' : $\omega \downarrow$

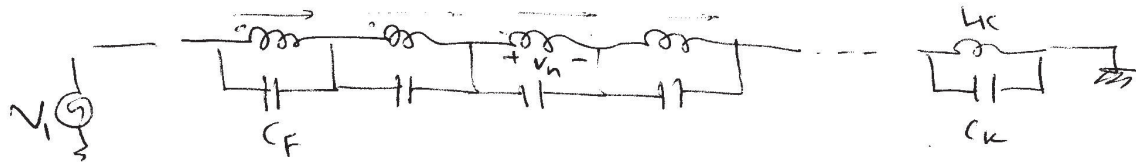
$$\therefore Q_b \approx \frac{\omega L}{R_s}, \quad Q_d = \frac{\omega L}{R_s}$$

if Q is dominant by R_p' : $\omega \uparrow$

$$\therefore Q_b \approx \frac{R_1}{\omega L}, \quad Q_d \approx \frac{2 R_1}{\omega L}$$

So the differential sol_n has higher Q @ high frequencies.

Problem 7.3r

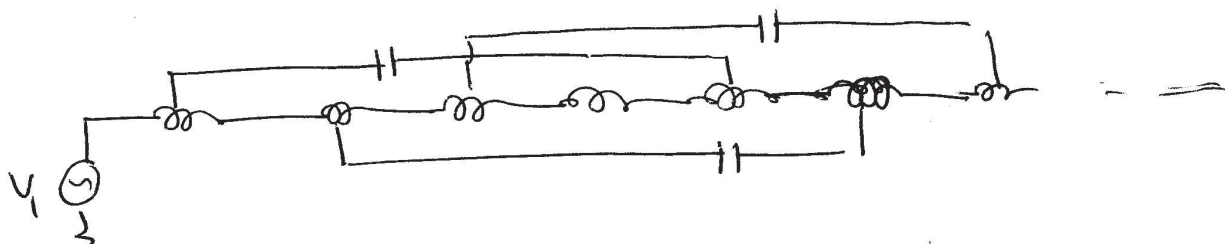


$$E_{tot} = K E_u = K \left[\frac{1}{2} C_F \left(\frac{1}{K} V_1 \right)^2 \right]$$

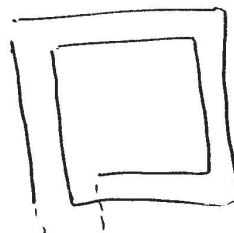
$$V_n \approx \tilde{I}_n j\omega L_n + j\omega \sum_{m=1}^{n-1} M_{nm} \tilde{I}_m + j\omega \sum_{m=n+1}^K M_{nm} \tilde{I}_m \approx j\omega \sum_{m=1}^K I_m L_m$$

$$\Rightarrow V_n^2 = \left(\frac{1}{K} \right)^2 V_1^2$$

$$\Rightarrow E_{tot} = \frac{1}{2K} C_F V_1^2 \rightarrow \text{but } C_F = \frac{C_{tot}}{K}$$



Total Number of segment in $L = 4N$
& total number of caps = $4(N-1)$.



$$L = \frac{L_{tot}}{4N} \quad \& \quad C_i = \text{average capacitance} = \frac{C_1 + C_2 + C_3 + \dots + C_{4(N-1)}}{4(N-1)}$$

$$E_{tot} = \sum_{i=1}^{4(N-1)} \frac{1}{2} C_i V_i^2$$

$$\approx \frac{1}{2} \left(\frac{V_1}{N} \right)^2 \sum_{i=1}^{4(N-1)} C_i$$

$$V_u = \frac{V_1}{4N} \text{ due to symmetry.}$$

$$K = 4N$$

$$V_i = 4 \frac{V_1}{4N} \approx \frac{V_1}{N}$$

$$C_q \equiv \frac{C_1 + C_2 + \dots + C_{4(N-1)}}{N^2} \rightarrow \text{Correct.}$$

3

Another Soln:-

$$C_{eq} = \frac{1}{K} C_F$$

$$= \frac{1}{4(N-1)} \times \frac{C_1 + C_2 + C_3 + \dots + C_{4(N-1)}}{4(N-1)}$$

$$C_{eq} = \frac{C_1 + C_2 + C_3 + \dots + C_{4(N-1)}}{16(N-1)^2}$$

↳ This soln didn't consider that the voltage between capacitance is $\frac{4V_1}{K}$ as it is between 4 inductors. That is why there is difference between both results.

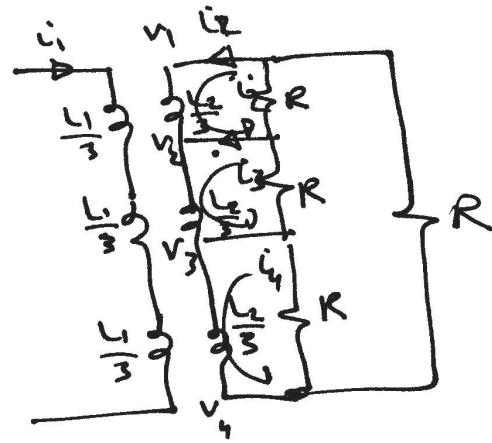
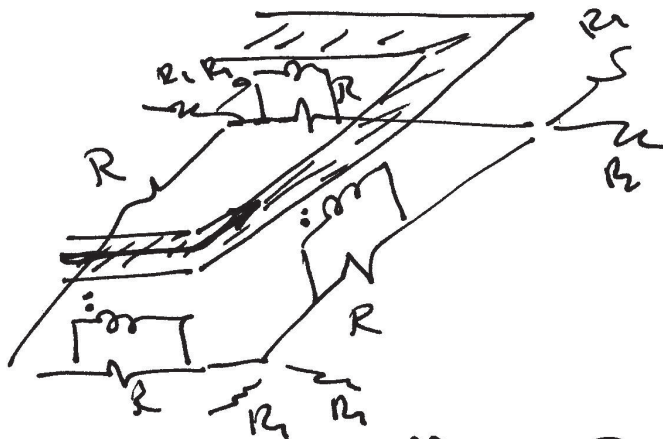
at $N=3$:

$$C_{eq} = \frac{C_1 + C_2 + \dots + C_8}{9} \Rightarrow \text{Correct answer.}$$

$$C_{eq} \approx \frac{C_1 + C_2 + \dots + C_8}{64} \Rightarrow \text{What is in the book.}$$

3'

Problem 7-7:



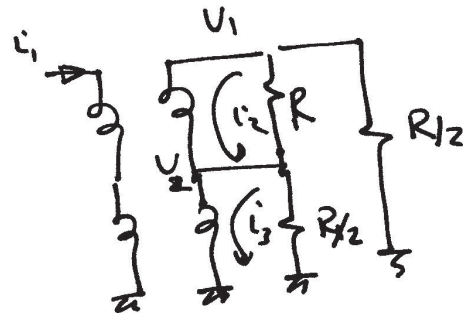
Neglecting Effect of R_2 :

$$V_{in} = i_1 L_1 s + \left(\frac{i_2 + i_3 + i_4}{3} \right) sM$$

due to odd symmetry: $i_2 = i_4$ & $V_1 = -V_4$ & $V_2 = -V_3$.

$$V_1 - V_2 = s i_2 L_2 / 3 + i_1 \frac{M}{3} s$$

$$V_2 = s i_3 L_2 / 6 + i_1 \frac{M}{6} s.$$



$$V_1 \left(\frac{2}{R} \right) + (V_1 - V_2) \frac{1}{R} + i_2 = 0.$$

$$V_2 \left(\frac{2}{R} \right) + (V_2 - V_1) \frac{1}{R} + i_3 = i_2.$$

$$V_1 = i_1 \frac{M}{2} s + s L_2 \left[\frac{i_2}{3} + \frac{i_3}{6} \right].$$

$$\Rightarrow \left[i_1 \frac{M}{2} s + s L_2 \left(\frac{i_3}{6} + \frac{i_2}{3} \right) \right] \frac{2}{R} + \frac{1}{R} (s i_2 L_2 / 3 + i_1 \frac{M}{3} s) + i_2 = 0.$$

$$\Rightarrow \left[i_1 \frac{M}{6} s + s L_2 \frac{i_3}{6} \right] \frac{2}{R} + \frac{1}{R} (s i_2 L_2 / 3 + i_1 \frac{M}{3} s) + i_3 = i_2.$$

from the 2 eqn get i_2 & i_3 as function of i_1

$$i_2 \left[1 + \frac{sL_2}{3} + \frac{2}{R} + \frac{sL_2}{3} \frac{1}{R} \right] + i_3 \left[\frac{sL_2}{3R} \right] = -i_1 \left[\frac{Ms}{R} + \frac{Ms}{3R} \right]$$

$$i_2 \left(1 + \frac{sL_2}{R} \right) + i_3 \left(\frac{sL_2}{3R} \right) = -i_1 \frac{4sM}{3R}$$

$$i_2 \left[-1 + \cancel{\frac{sL_2}{3R}} - \frac{sL_2}{3R} \right] + i_3 \left[\frac{sL_2}{3R} + 1 \right] = -i_1 \left[\frac{sM}{3R} + \frac{sM}{3R} \right] = 0$$

$$\Rightarrow \boxed{i_3 = i_2} \quad \Rightarrow i_2 \left(1 + \frac{4sL_2}{3R} \right) = -i_1 \frac{4sM}{3R}$$

$$\boxed{i_2 = -i_1 \left(\frac{4sM}{3R + 4sL_2} \right)}$$

$$\Rightarrow V_{in} = i_1 L_1 s + i_2 sM = i_1 L_1 s - i_1 \frac{4sM}{3R + 4sL_2} sM$$

$$\Rightarrow Y_{11} = \frac{i_1}{V_{in}} = \frac{1}{sL_1 - \frac{4sM}{(3R + 4sL_2)} sM}$$

$$= \frac{3R + 4sL_2}{3R sL_1 + 4s^2 L_1 L_2 - 4s^2 M^2}$$

$$Y_{11} = \frac{3R + 4sL_2}{3R sL_1 + 4s^2 (1 - k^2) L_1 L_2} \quad \text{if } k=1$$

$$\Rightarrow Y_{11} = \frac{1}{sL_1} + \frac{4}{3R} \times \left(\frac{L_2}{L_1} \right) \rightarrow \text{transformer ratio.} \quad \boxed{R_{sub} \approx \frac{3R}{4}}$$

5

$$V_1 - V_2 = s i_2 L_2 / 3 + s i_1 M / 3.$$

$$V_2 = s i_3 L_2 / 6 + s i_1 M / 6.$$

$$\Rightarrow V_1 \left(\frac{2}{R_x} \right) + (V_1 - V_2) \frac{1}{R} + i_2 = 0.$$

$$V_2 \left(\frac{2}{R_x} \right) + (V_2 - V_1) \frac{1}{R} + i_3 = i_2.$$

$$\Rightarrow V_1 = i_1 \frac{M}{2} s + s L_2 \left(\frac{i_2}{3} + \frac{i_3}{6} \right).$$

$$\Rightarrow \left[i_1 \frac{M}{2} s + s L_2 \left(\frac{i_3}{6} + \frac{i_2}{3} \right) \right] \left(\frac{2}{R_x} \right) + \frac{1}{R} \left[s i_2 \frac{L_2}{3} + i_1 \frac{M}{3} s \right] + i_2 = 0$$

$$\Rightarrow \left[i_1 \frac{M}{6} s + s L_2 \left(\frac{i_3}{6} \right) \right] \frac{2}{R_x} - \frac{1}{R} \left(s i_2 \frac{L_2}{3} + i_1 \frac{M}{3} s \right) + i_3 = i_2.$$

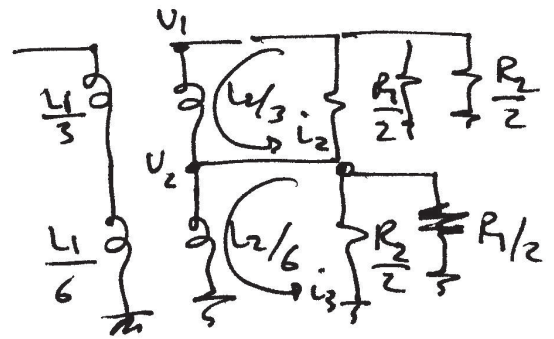
$$\Rightarrow i_2 \left[1 + \frac{s L_2}{3} \frac{2}{R_x} + \frac{s L_2}{3 R} \right] + i_3 \left[\frac{s L_2}{3 R_x} \right] = -i \left[\frac{M s}{R_x} + \frac{M s}{3 R} \right].$$

$$\Rightarrow i_2 \left(\frac{s L_2}{3} \right) \left(\frac{2}{R_x} + \frac{1}{R} \right) + i_3 \left(\frac{s L_2}{3 R_x} \right) = -i \left[\frac{M s}{1} \right] \left(\frac{1}{R_x} + \frac{1}{3 R} \right). \text{ eqn 1}$$

$$\& - \left[1 + \frac{s L_2}{3 R} \right] i_2 + \left[1 + \frac{s L_2}{3 R_x} \right] i_3 = -i_1 \left[\frac{M s}{3 R_x} - \frac{M s}{3 R} \right] - i \frac{M s}{3 R_2}. \text{ eqn 2}$$

from ① & ② get i_2 & i_3 as function in i_1 :-

$$\Rightarrow \text{get } V_{in} \text{ as function of } i_1 \Rightarrow Y_{11} = \frac{i_1}{V_{in}}.$$

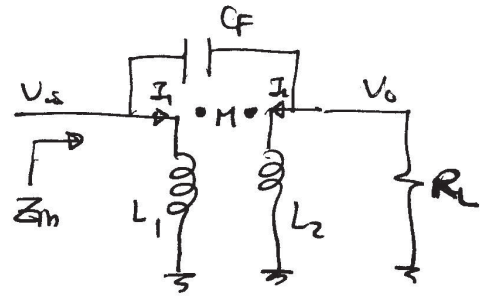


$$R_x = R_1 \parallel R_2$$

$$R = R_1$$

Problem 7.9.

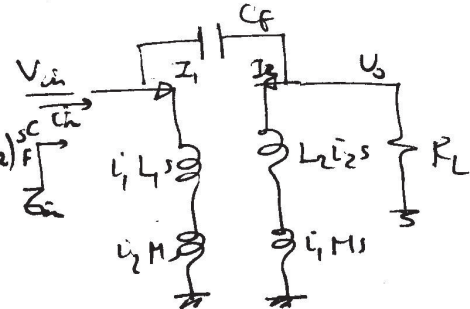
$$\frac{V_o}{R_L} + (V_o - V_m) s C_F + I_2 = 0.$$



$$V_o = s L_2 i_2 + s M i_1.$$

$$V_m = s L_1 i_1 + s M i_2.$$

$$\rightarrow \frac{s L_2 i_2 + s M i_1}{R_L} + (L_2 i_2 - M i_1) s C_F + I_2 = 0.$$



$$I_2 \left[1 + \frac{s L_2}{R_L} + s (L_2 - M) s C_F \right] = -I_1 \left[\frac{s M}{R_L} + s (M - L_1) s C_F \right].$$

$$\begin{aligned} V_m &= s L_1 i_1 + s M i_2 \\ &= s L_1 i_1 - \frac{\left[\frac{s M}{R_L} + s^2 (M - L_1) C_F \right]}{\left[1 + \frac{s L_2}{R_L} + s^2 (L_2 - M) C_F \right]} s M i_1 \end{aligned}$$

$$\begin{aligned} (V_m - V_o) s C_F &= [s L_1 I_1 - s M I_1 + s M I_2 - s L_2 I_2] s C_F \\ &= s \left[(L_1 - M) I_1 + (L_2 - M) \frac{s M / R_L + s^2 (M - L_1) C_F}{1 + \frac{s L_2}{R_L} + s^2 (L_2 - M) C_F} I_1 \right] C_F \end{aligned}$$

$$Z_{in}^{(b)} = \frac{V_m}{I_1 + (V_m - V_o) s C_F} = \frac{s L_1 - s M \frac{s M / R_L + s^2 (M - L_1) C_F}{1 + \frac{s L_2}{R_L} + s^2 (L_2 - M) C_F}}{1 + s^2 (L_1 - M) C_F + s^2 C_F (L_2 - M) \frac{s M / R_L + s^2 (M - L_1) C_F}{1 + \frac{s L_2}{R_L} + s^2 (L_2 - M) C_F}}.$$

$$= \frac{s L_1 \left(1 + \frac{s L_2}{R_L} + s^2 (L_2 - M) C_F \right) - s M \left(\frac{s M}{R_L} + s^2 (M - L_1) C_F \right)}{\left(1 + s^2 (L_1 - M) C_F \right) \left(1 + \frac{s L_2}{R_L} + s^2 (L_2 - M) C_F \right) + s^2 (L_2 - M) C_F \left(\frac{s M}{R_L} + s^2 (M - L_1) C_F \right)}$$

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$$Z_m(s) = \frac{sL_1 + \frac{s^2 L_1 L_2}{R_L} - \frac{s^2 M^2}{R_L} + \cancel{s^3} s^3 C_F (L_1 L_2 - L_1 M - M^2 + L_2 M)}{1 + \frac{sL_2}{R_L} + s^2(L_2 - M)C_F + s^2(L_1 - M)C_F + \frac{sL_2}{R_L} s^2(L_1 - M)C_F + s^3 C_F^2 (L_1 - M)(L_2 - M) + s^2(L_2 - M) \frac{sM}{R_L} C_F + s^4(L_2 - M)(L_1 - M)C_F^2}$$

$$Z_m(s) = \frac{sL_1 + \frac{s^2}{R_L} (L_1 L_2 - M^2) + s^3 C_F (L_1 L_2 - M^2)}{1 + \frac{sL_2}{R_L} + s^2 C_F (L_1 + L_2 - 2M) + \frac{s^3 C_F}{R_L} [L_2 M - M^2 + L_1 L_2 - L_2 M]} \\ = \frac{sL_1 + \frac{s^2 L_1 L_2}{R_L} (1 - k^2) + s^3 C_F L_1 L_2 (1 - k^2)}{1 + \frac{sL_2}{R_L} + \frac{s^3 C_F L_1 L_2}{R_L} (1 - k^2) + s^2 C_F [L_1 + L_2 - 2k\sqrt{L_1 L_2}]}$$

assuming $C_F = 0$

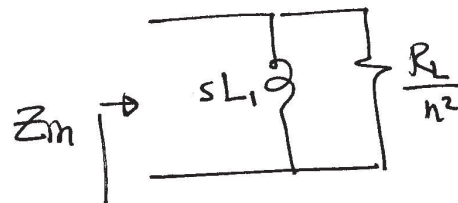
$$\therefore Z_m(s) = \frac{1 + \frac{sL_2}{R_L} (1 - k^2)}{1 + \frac{sL_2}{R_L}} sL_1$$

at $k=1$:

$$Z_m(s) = \frac{1}{1 + \frac{sL_2}{R_L}} sL_1$$

$$Y_m(s) = \left(1 + \frac{sL_2}{R_L}\right) \frac{1}{sL_1} = \frac{1}{sL_1} + \frac{L_2}{L_1} \frac{1}{R_L}, \quad n = \sqrt{\frac{L_2}{L_1}}$$

$$Y_m(s) = \frac{1}{sL_1} + \frac{n^2}{R_L}$$



8

Problem 7.10c

	Metal 8	Metal 7	Metal 6	Metal 5	Subs.
Metal 9	52	16	12	9.5	4.4
Metal 8		52	24	16	5.4
Metal 7			88	28	6.1
Metal 6				88	7.1
Metal 5					8.6

For a single metal layer inductor (metal-9)

$$W = 4 \mu\text{m}, S = 2 \mu\text{m}, N = 4, l_{\text{tot}} = 2000 \mu\text{m}.$$

$$\therefore L = 1.3 \times 10^{-7} \frac{l_{\text{tot}}^{5/3}}{\left[\frac{l_{\text{tot}}}{4N} + W + (N-1)(W+S) \right]^{1/3} W^{0.083} (W+S)^{0.25}}$$

$$L = 4.45 \text{ nH}.$$

Capacitance in this case is $C_{\text{tot}} = \text{Area (M9)} \times 44 \text{ aF}/\mu\text{m}^2$
 $= 8000 \times 4.4 \text{ aF} = 35.2 \text{ fF}$

$$C_{\text{eq}} = \frac{1}{3} \times C_{\text{tot}} = 11.73 \text{ fF}.$$

For stacked four layer inductor:-

choosing M9, M8, M7, M5 to have minimum capacitance.

$$\text{Inductance} \approx L \times 16 = 4.45 \text{ nH}$$

$$\therefore \boxed{L = 0.278 \text{ nH}}$$

Using $N=1$, $W=4 \mu\text{m}$, $S=2 \mu\text{m}$, $L=0.278 \text{ nH}$

$$l_{bt}^{5/3} = (2138.5) \left[\frac{l_{bt}}{4} + W \right]^{1/3} (W)^{0.083} (W+S)^{0.25}$$

$$l_{bt}^{5/3} = 3755 \left(\frac{l_{bt}}{4} + W \right)^{1/3} \longrightarrow l_{bt} = 343 \mu\text{m}$$

$$l_{bt} = (3755)^{3/5} \left[\frac{l_{bt}}{4} + W \right]^{1/5}$$

Using $N=2$

$$l_{bt}^{5/3} = (2138.5) \left[\frac{l_{bt}}{8} + W + (W+S) \right]^{1/3} W^{0.083} (W+S)^{0.25}$$

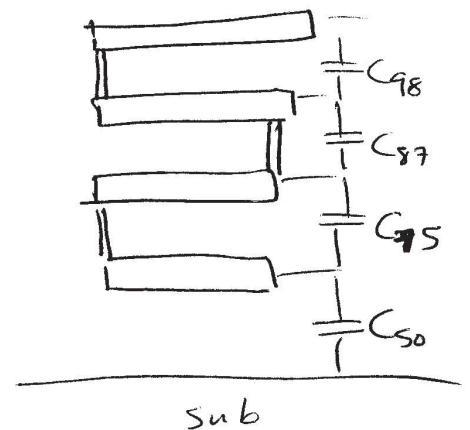
$$l_{bt} = (3755)^{3/5} \left[\frac{l_{bt}}{8} + 2W + S \right]^{1/5} \longrightarrow l_{bt} = 302.4 \mu\text{m}$$

$$\Rightarrow C_{98} = 302.4 \times 4 \times 52 = 62.9 \text{ fF}$$

$$C_{87} = 302.4 \times 4 \times 52 = 62.9 \text{ fF}$$

$$C_{75} = 302.4 \times 4 \times 28 = 33.9 \text{ fF}$$

$$C_{50} = 302.4 \times 4 \times 8.6 = 10.4 \text{ fF}$$



from Eq. (7.81)

$$C_{eq} = \frac{4 \sum_{m=1}^{n-1} C_m + C_{sub}}{3n^2}$$

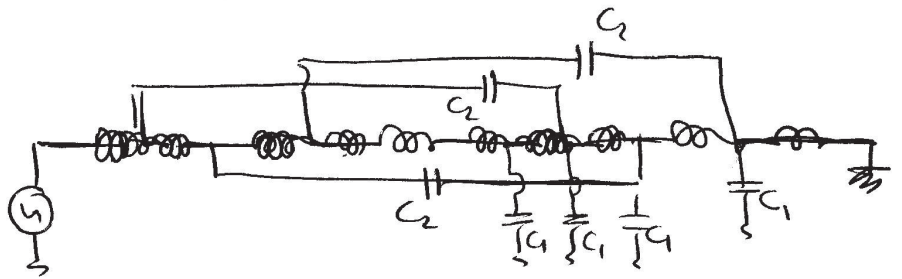
$$C_{eq} = \frac{4 [62.9 \times 2 + 33.9] + 10.4}{3 \times 4^2}$$

$$= 13.53 \text{ fF}$$

$$\Rightarrow \text{Corrected one}$$

$$C_{eq} = \frac{3 \sum_{m=1}^{n-1} C_m + C_{sub}}{3n^2}$$

$$\Rightarrow C_{eq} = 10.20 \text{ fF}$$



$$E_{n_{sub}} \cong \frac{1}{2} C_1 \left(\frac{n}{2k} \right)^2 V_1^2 = \frac{1}{8} C_1 \left(\frac{n^2}{k^2} \right) V_1^2.$$

$$E_{n_2} \cong \frac{1}{2} C_2 \left[\frac{(n+k) - n}{2k} \right]^2 V_1^2 = \frac{1}{8} C_2 V_1^2.$$

$$E_{tot} = \sum_{n=0}^k E_{n_2} + \sum_{n=1}^k E_{n_{sub}} = \frac{k}{8} C_2 V_1^2 + \frac{1}{8} C_1 V_1^2 \sum_{n=1}^k \frac{n^2}{k^2}$$

$$= \frac{C_{2tot}}{8} V_1^2 + \frac{1}{8} C_1 V_1^2 \frac{(k+1)(2k+1)}{6k}$$

$$C_{eq} = \frac{1}{V_1^2} \frac{2E_{tot}}{3} = \frac{C_{2tot}}{4} + \frac{C_{1tot}}{12} \quad \text{for } \lim_{k \rightarrow \infty}$$

$$= \frac{3 C_{1tot} + C_{2tot}}{12}.$$

$$C_{eq} = \frac{3 \sum_{m=1}^{n-1} C_m + C_{sub_{tot}}}{3n^2}$$

$n =$ is the number of stacked layers.